

Problem Set 1

Applied Stats/Quant Methods 1

Due: October 8, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in R, please include the code you used to get your answers. Please also include the .R file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday October 8, 2026. No late assignments will be accepted.

Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

IQ Scores:

```
1 y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,  
      80, 97, 95, 111, 114, 89, 95, 126, 98)
```

Number of Observations:

```
1 n = length(y)
```

1. Find a 90% confidence interval for the average student IQ in the school (As a hint, you first need the mean and SD to create a CI):

Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

Descriptive Statistics:

```

1 mean_y <- sum(y)/n
2 sd_y <- sqrt(sum((y - mean_y) ^ 2) / (n - 1))
3 se_y <- sd_y/sqrt(n)

```

Critical Value:

```

1 t_critical <- qt(0.95, df = n - 1)

```

90% Confidence Interval:

```

1 ci_90_lower = mean_y - t_critical * se_y
2 ci_90_upper = mean_y + t_critical * se_y

```

Analysis:

By definition of the confidence interval formula (confidence interval = sample statistic +/- critical value*standard of error), we need to identify our sample statistic, critical value, and margin of error. To calculate the sample statistic, we use the mean of all the observations to identify the trends of all the IQ scores on average. This also makes sense because the data is continuous and relatively normally distributed as shown in the visualisation below. Then, for the critical value, a t-critical value would be more suitable than a z-critical value because the sample size of IQ Scores is less than 30 as we observe using the length function (n=25). To find the t-critical value, we can use the built-in qt() function in R. However, it is important to note that for a 90% confidence interval, the appropriate probability parameter is 0.95 and not 0.90. This follows the idea that we are looking at both tails of the distribution, so when only observing one tail, the area of 0.10 decreases down to 0.05, which translates into a parameter of 0.95. Given that both the population mean and population standard deviation is unknown, we need to subtract one degree from the sample size to receive the degree of freedoms in the second parameter. Then, we need to solve for the standard deviation and square this value to receive the standard error. With both the critical value and standard of error now, we can calculate the margin of error by multiplying the two values. Following the formula for the confidence interval by definition, we can use our information of the sample statistic (sample mean) and add the margin of error to receive the upper bound of the 90% confidence interval. Then, we subtract the margin of error from the sample statistic to then receive the lower bound of the 90% confidence interval.

We result in the confidence interval of (93.9599, 102.9201).

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

i. Assumptions (Visualising the Data):

```

1 pdf("plot_1.pdf")
2
3 hist(y,
4       breaks = 8,
5       probability = TRUE,

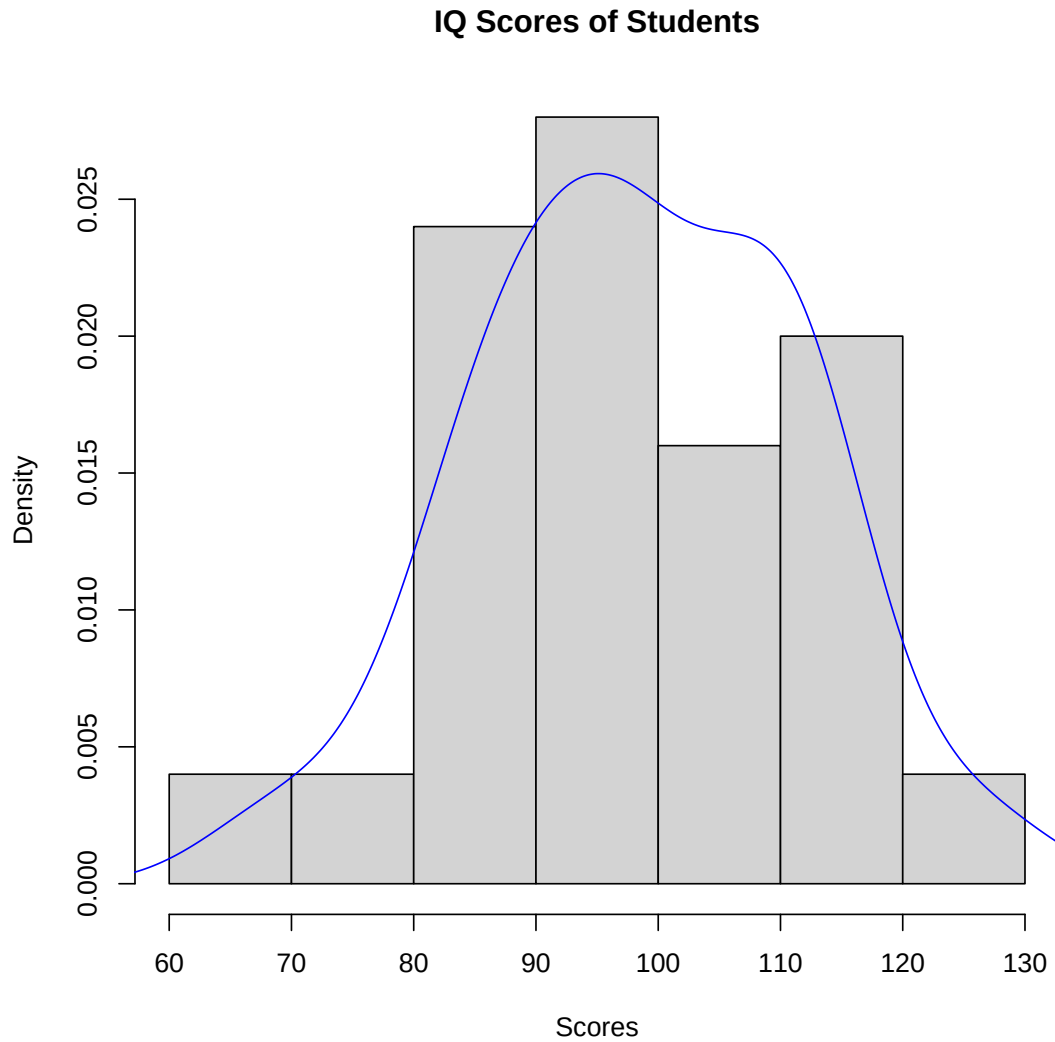
```

```

6     main = "IQ Scores of Students",
7     xlab = "Scores")
8
9 lines(density(y), col = "blue")
10
11 dev.off()

```

Figure 1: IQ Scores of Students



ii. *Null and Alternate Hypotheses:*

Question: Is the average IQ score in the counselor's school higher than the average IQ score of 100 among all schools in the country?

iii. Hypotheses:

$$\begin{aligned} H_0 : \mu &= 100 && \text{(the average IQ score is 100)} \\ H_A : \mu &> 100 && \text{(the average IQ score is greater than 100)} \end{aligned}$$

iv. Critical Value:

```
1 mu_0 = 100
1 t_critical <- qt(0.95, df = n - 1)
```

iv. Test Statistic:

```
1 t_score <- (mean_y - mu_0) / se_y
```

iv. P-value:

```
1 p_value <- pt(t_score, df = n-1, lower.tail = FALSE)
```

Check Using the Built-in `t.test()` Function:

```
1 t.test(y, mu = 100, alternative = "greater", conf.level = 0.95)
```

v. Conclusion:

Analysis:

To figure out whether the average student IQ in the counselor's school is higher than the average IQ score among all schools in the country, we proceed with a one-tailed significance test. Following the 5 steps of hypothesis testing, we first identify whether the sample meets the assumption criterias. We know that IQ is a continous variable, measured on an interval scale (contains a true zero), and that the sample size is 25. Information given to us include that the sample is random. The visualisation of the sample distribution below also illustrates that the data relatively follows a normal distribution from its bell-shaped curve. This is important to emphasize because the sample size is less than 30, so we cannot use the central limit theorem and a large amont of sampling instances to determine normality. Lastly, we can assume that the IQ scores in the sample are independent of each other because they are drawn from a school population, where most students are not from the same family such that genetic similarities would cause dependency across observations.

After verifying that all the assumptions are met, we formulate the null and alternate hypothesis. The null hypothesis states that there is no difference between the IQ score within the counselor's school among all schools in the country, such that both on average is 100. In contrast, the alternate hypothesis states that the average IQ score is greater than that of students in all schools in the country.

There are two ways we can come to the right conclusion. The first method would be to calculate the t-critical value and t-statistic. Again, the t-test is used because the sample size is less than 30 and both the population mean and standard deviation are unknown. For the alpha threshold of 0.05, we enter 0.95 as the probability in the

first parameter of the built-in `qt()` function to calculate the t-critical value since we are only observing a one-tail test. If the t-statistic is significantly greater than the t-critical value, then the difference is highly unlikely to be a random occurrence, allowing us to reject the null hypothesis. However, given that the t-statistic value is -0.59574 and the t-critical value is 1.71088, the t-statistic value is significantly lower than the t-critical value. This indicates that any differences within the sample could occur from random chance, so we fail to reject the null hypothesis. The p-value corroborates this conclusion, but simply provides a second strategy to approach the significance test. The p-value we observe is 0.7215, which is significantly greater than the alpha threshold of 0.05. Hence, this indicates that in a world where the null is true, it is unlikely that we observe a value more extreme than the observed value. With that in mind, we would reject the null hypothesis.

We fail to reject the null hypothesis and can conclude that there is not sufficient evidence to say that the average student IQ in the counselor's school is higher than the average IQ score (100) among all the schools in the country.

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

State	50 states in US
Y	per capita expenditure on shelters/housing assistance in state
X1	per capita personal income in state
X2	Number of residents per 100,000 that are "financially insecure" in state
X3	Number of people per thousand residing in urban areas in state
Region	1=Northeast, 2= North Central, 3= South, 4=West

Explore the `expenditure` data set and import data into R.

Import Data:

```
1 expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2026/main/datasets/expenditure.txt", header=T)
```

Explore Data:

```
1 head(expenditure)
2 str(expenditure)
3 summary(expenditure)
```

- Please plot the relationships among *Y*, *X1*, *X2*, and *X3*? What are the correlations among them (you just need to describe the graph and the relationships among them)?

Define Variables:

```
1 vars <- c("Y", "X1", "X2", "X3")
```

Variable Labels:

```
1 labels <- c(
2   "Expenditure",
3   "Income",
4   "Financially Insecure",
5   "Urban"
6 )
```

Set Up Plot:

```
1 pdf("plot_2.pdf", width = 21, height = 12)
2
3 par(mfrow = c(2, 3), cex = 1.8)
```

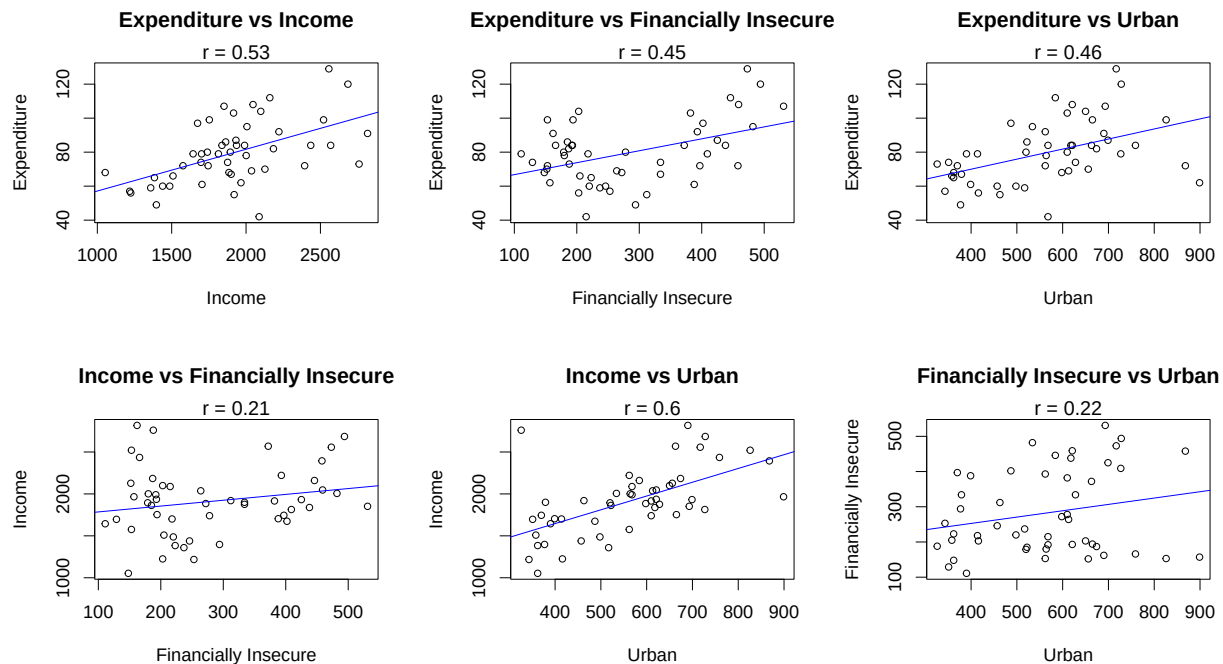
Pairwise Scatterplots and Correlations:

```

1 for (i in 1:(length(vars) - 1)) {
2   for (j in (i + 1):length(vars)) {
3
4     plot(
5       expenditure[[vars[j]]],
6       expenditure[[vars[i]]],
7       xlab = labels[j],
8       ylab = labels[i],
9       main = paste(labels[i], "vs", labels[j]),
10      cex = 0.8
11    )
12
13    abline(
14      lm(expenditure[[vars[i]]] ~ expenditure[[vars[j]]]),
15      col = "blue",
16    )
17
18    r <- cor(
19      expenditure[[vars[i]]],
20      expenditure[[vars[j]]]
21    )
22
23    mtext(
24      paste("r =", round(r, 2)),
25      cex = 2
26    )
27  }
28 }
29
30 dev.off()

```

Figure 2: Pairwise Relationships and Correlations



Analysis:

- Please plot the relationship between Y and $Region$? On average, which region has the highest per capita expenditure on housing assistance?

Define Regions:

```
1 regions <- factor(
2   expenditure$Region,
3   levels = c(1, 2, 3, 4),
4   labels = c("North East", "North Central", "South", "West")
5 )
```

Compare Regional Means and Medians:

```
1 for (i in levels(regions)) {
2   region_mean <- round(mean(expenditure$Y[regions == i]), 2)
3   region_median <- round(median(expenditure$Y[regions == i]), 2)
4
5   print(paste(i, "mean expenditure =", region_mean))
6   print(paste(i, "median expenditure =", region_median))
7   cat("\n")
8 }
```

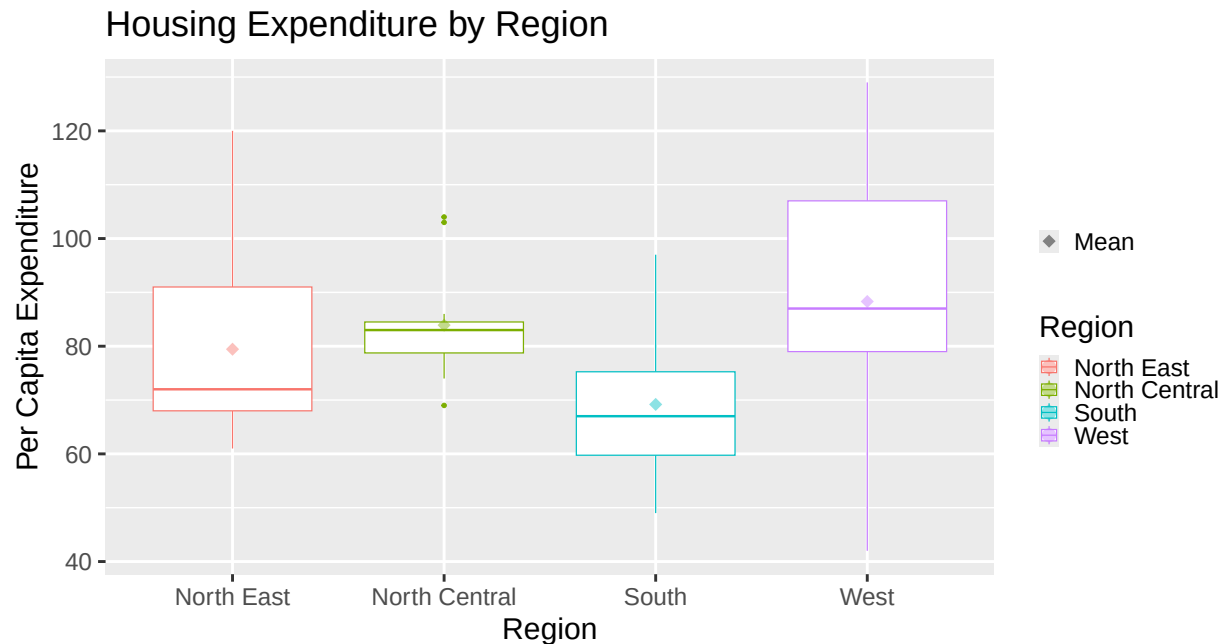
Plot Housing Expenditure by Region:

```

1 plot_3 <- ggplot(expenditure, aes(x = regions, y = Y, color = factor(
2   regions))) +
3   geom_boxplot() +
4   stat_summary(
5     aes(shape = "Mean"),
6     fun = mean,
7     geom = "point",
8     size = 5,
9     alpha = 0.45
10  ) +
11  labs(
12    title = "Housing Expenditure by Region",
13    x = "Region",
14    y = "Per Capita Expenditure",
15    color = "Region",
16    shape = ""
17  ) +
18  scale_shape_manual(
19    values = c("Mean" = 18)
20  ) +
21  theme_gray(base_size = 26)
22 ggsave("plot_3.pdf", plot = plot_3, width = 15, height = 8)

```

Figure 3: Housing Expenditure by Region



Analysis:

- Please plot the relationship between Y and $X1$? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

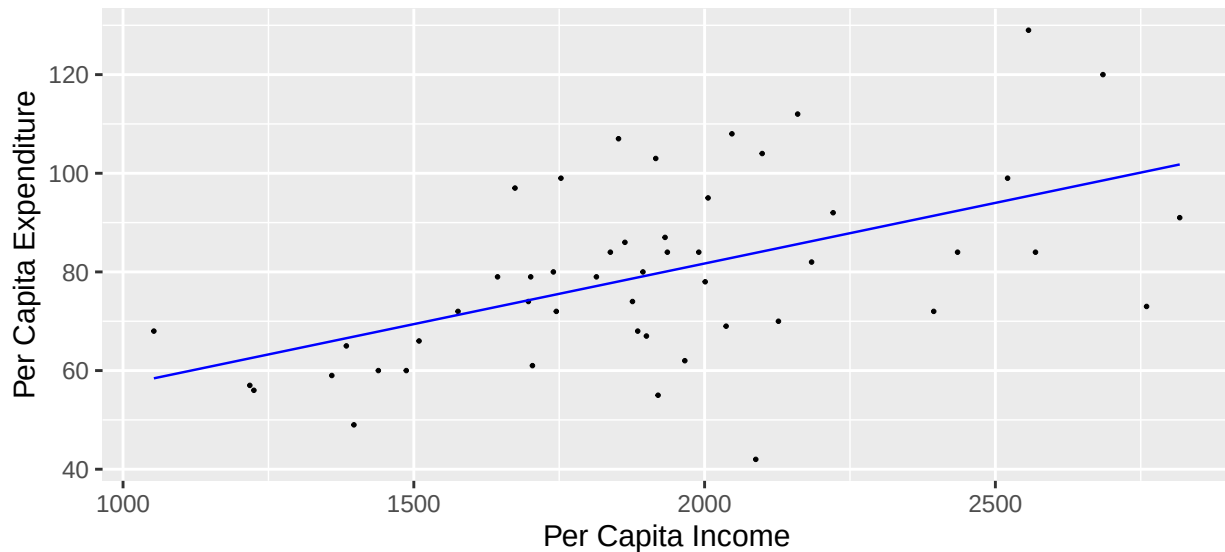
Housing Expenditure and Income:

```
1 plot_4 <- ggplot(expenditure, aes(x = X1, y = Y)) +
2   geom_point() +
3   geom_smooth(
4     method = "lm",
5     color = "blue",
6     se = FALSE
7   ) +
8   labs(
9     title = "The Relationship between Housing Expenditure and Income by
10    Region",
11    subtitle = paste("r =", round(cor(expenditure$X1, expenditure$Y), 3))
12    ,
13    x = "Per Capita Income",
14    y = "Per Capita Expenditure"
15  ) +
16  theme_gray(base_size = 26)
```

```
16 ggsave("plot_4.pdf", plot = plot_4, width = 15, height = 8)
```

Figure 4: Housing Expenditure versus Income

The Relationship between Housing Expenditure and Income by Region
 $r = 0.532$



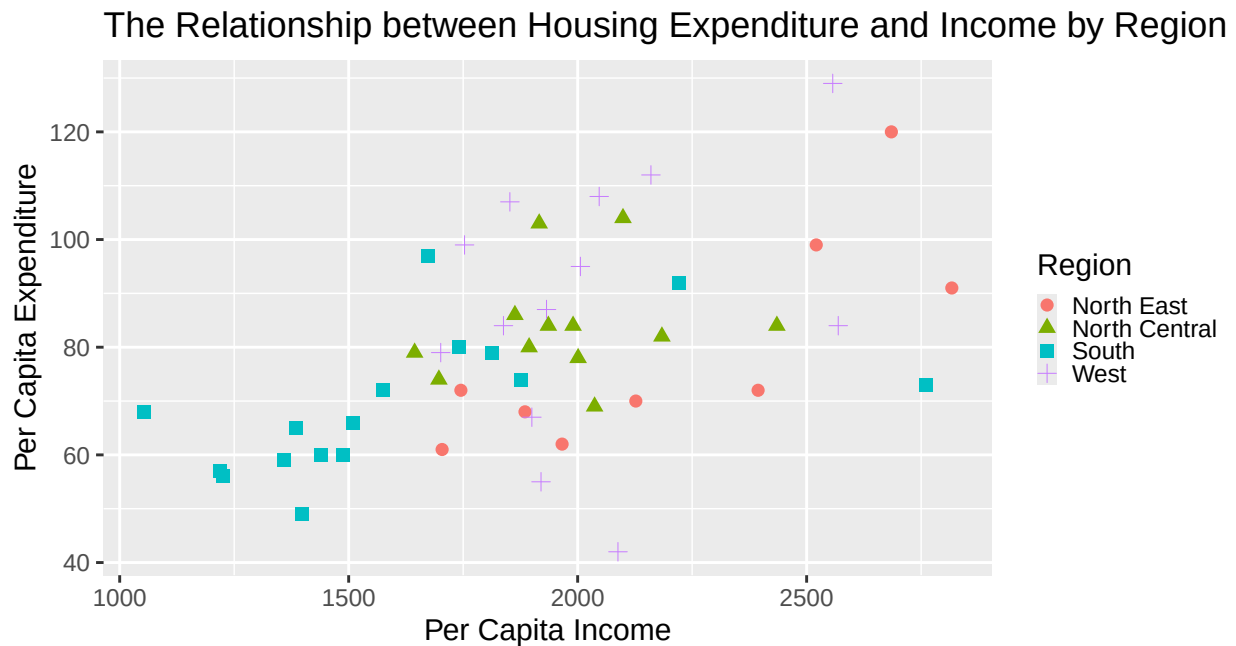
Housing Expenditure and Income by Region:

```

1 plot_5 <- ggplot(
2   expenditure ,
3   aes(
4     x = X1,
5     y = Y,
6     color = factor(regions),
7     shape = factor(regions)
8   )
9 ) +
10  geom_point(size = 5) +
11  labs(
12    title = "The Relationship between Housing Expenditure and Income by
13    Region",
14    x = "Per Capita Income",
15    y = "Per Capita Expenditure",
16    color = "Region",
17    shape = "Region"
18  ) +
19  theme_gray(base_size = 26)
20 ggsave("plot_5.pdf", plot = plot_5, width = 15, height = 8)

```

Figure 5: Housing Expenditure versus Income by Region



Correlations by Region:

```

1 for (i in levels(regions)) {
2   region_correlation <- round(cor(expenditure$X1[regions == i],
3     expenditure$Y[regions == i]), 3)

```

```
3  
4 print(paste(i, "correlation =", region_correlation))  
5 cat("\n")
```

Analysis: